

From Single Agent to Multi-Agent Systems

Non-stationarity · Taxonomy · Nash Equilibrium · PettingZoo · MPE

Physics hook: A single spin in an external field $\rightarrow$ interacting spin system. Mean-field theory is no longer sufficient.

Agenda -- 2 hours

20'	Recap: single-agent MDP
30'	What changes with multiple agents
20'	Taxonomy · Nash Equilibrium
30'	PettingZoo: AEC vs Parallel architecture
20'	First environment: <code>simple_spread</code> (MPE)

What we already know: single-agent MDP

MDP = $(\mathcal{S}, \mathcal{A}, P, R, \gamma) \rightarrow$ Policy $\pi : \mathcal{S} \rightarrow \mathcal{A} \rightarrow$ Bellman equation: optimal Q^*

Stationary environment

$P(s'|s, a)$ and $R(s, a)$ do not change over time. The agent can learn off-policy and use a replay buffer.

✓ assumption 1

Full observability

The Markov state captures all relevant information. Extension to POMDP: belief state.

✓ assumption 2

Single agent, single reward

$\max_{\pi} \mathbb{E}[\sum_t \gamma^t r_t]$ — well-defined optimisation with a unique solution (for $\gamma < 1$).

✓ assumption 3

$\Rightarrow$ All three assumptions break down when we introduce multiple agents.

What changes: the fundamental problem

Non-stationarity: the environment perceived by agent i depends on the policies of all other agents $j \neq i$. If others are learning, the environment shifts under every agent's feet — even if P itself is stationary.

1 — Bellman is no longer sufficient $Q^*(s, a)$ implicitly depends on others' policies. The solution is not unique: the space is a *game*, not an optimisation.

2 — Replay buffer becomes invalid In DQN, stored transitions were valid because P was stationary. With multiple learning agents, stale (s, a, r, s') tuples change distribution.

3 — Partial observability by default Each agent sees only a local window. The global state is inaccessible unless explicit communication is available.

The physicist's analogy

Single agent $\equiv$ Isolated spin

$$H = -h\sigma_i$$

- External field $h \rightarrow$ reward signal
- $\sigma_i \in \{\uparrow, \downarrow\} \rightarrow$ agent action
- Fixed Hamiltonian, unique solution

$\xrightarrow{N \rightarrow \infty}$

Multi-agent $\equiv$ Interacting spins

$$H = -J \sum_{ij} \sigma_i \sigma_j - h \sum_i \sigma_i$$

- Coupling $J_{ij} \rightarrow$ reward coupling
- Spin i feels the field of spins j
- Environment changes if others change

$\Rightarrow$ *Mean field overestimates cooperation: agent-agent correlations make all the difference.*

Taxonomy of multi-agent environments

Cooperative

shared reward: $r_i = r$ for all i

Examples:

Cooperative Navigation (MPE)
Distributed traffic control
Robot swarm coordination

[Physics]: Ferromagnetic spin system: all spins want to align. Min energy = max cooperation.

Competitive

zero-sum: $\sum_i r_i = 0$

Examples:

Go, Chess (AlphaZero)
Simultaneous auctions
Predator-prey

[Physics]: Antiferromagnetic system: neighbours oppose each other. Frustration $\leftrightarrow$ mixed equilibria.

Mixed

individual rewards + common good

Examples:

Traffic: each car vs. global flow
Financial markets
Multi-team systems

[Physics]: Frustrated system with both cooperation and competition. Spin glasses as limiting case.

Nash Equilibrium -- the intuition

Nash Equilibrium: a strategy profile $(\pi_1^*, \pi_2^*, \dots, \pi_N^*)$ such that no agent can increase its own reward by *unilaterally* deviating, while all others keep their policy fixed.

RPS -- payoff matrix

	Rock	Paper	Scissors
Rock	0,0	-1,+1	+1,-1
Paper	+1,-1	0,0	-1,+1
Scissors	-1,+1	+1,-1	0,0

NE of RPS

No pure strategy is a NE:

→ if I always play Rock, opponent plays Paper

→ no pure profile is stable

Unique mixed strategy:

$\pi^* = (1/3, 1/3, 1/3)$ for both
EU = 0 — no one can improve.

Nash 1950: Every finite game has
> 1 NE.

Prisoner's Dilemma -- Nash $\neq$ Social Optimum

Payoff matrix (years in prison,
less = better)

	B: Cooperate	B: Defect
A: Cooperate	-1, -1	-3, 0
A: Defect	0, -3	-2, -2

→ Social optimum

→ Nash Equilibrium

Lessons for MAS design

NE $\neq$ Pareto optimum

Rational equilibrium yields $(-2, -2)$ instead of $(-1, -1)$. Individual rationality $\neq$ collective welfare.

Reward shaping

We can push the NE towards the social optimum by redesigning the reward structure → reward shaping & incentive mechanisms.

Repetition & cooperation

Repeated game (Folk Theorem): cooperation can emerge as a NE.

PettingZoo -- the Gymnasium for MARL

PettingZoo (Terry et al., 2021) is the standard library for MARL environments. Two main APIs: **AEC** (Agent Environment Cycle) and **Parallel**. Same philosophy as Gymnasium: unified interface for dozens of environments.

AEC -- Agent Environment Cycle

Sequential turns: one agent at a time.

```
env.reset()
for agent in env.agent_iter():
    obs, rew, term,
    trunc, info = env.last()
    action = policy(obs)
    env.step(action)
```

Parallel -- Simultaneous steps

All agents act at the same time.

```
obs, _ = env.reset()
while env.agents:
    actions = {a: policy(obs[a])
               for a in env.agents}
    obs, rew, term,
    trunc, info = env.step(actions)
```

API Comparison: Gymnasium → PettingZoo

Concept	Gymnasium (single)	PettingZoo Parallel	Note
Observation	<code>obs = env.reset()</code>	<code>obs_dict = env.reset()</code>	dict keyed by agent id
Action space	<code>env.action_space</code>	<code>env.action_space(a)</code>	per-agent, may differ
Step	<code>obs, r, term, trunc, info = env.step(a)</code>	<code>obs, rew, ... = env.step(actions_d</code>	actions dict → obs dict
Reward	<code>r: float</code>	<code>rew: dict[agent→float]</code>	coop: sum; comp: per-agent
Done	<code>term</code> or <code>trunc</code>	<code>all(term.values())</code>	or per-agent (elimination)
Agent list	-- (unique)	<code>env.agents: list</code>	may change (elimination)

Migrating from Gymnasium to PettingZoo = adding one dict layer. The training logic stays the same.

First MAS environment: simple_spread

Multi-Particle Environment (MPE)

Agents N continuous agents in 2D space

Landmarks N fixed target positions

Goal Each agent covers a different landmark

Reward $r = -\min_{\ell} d(a_i, \ell)$ for uncovered ℓ

Install & quickstart

```
pip install pettingzoo[mpe]
pip install supersuit
```

```
from pettingzoo.mpe import \
    simple_spread_v3
env = simple_spread_v3.parallel_env(
    N=3, max_cycles=25,
    continuous_actions=False)
obs, infos = env.reset()
while env.agents:
    actions = {
        a: env.action_space(a).sample()
        for a in env.agents}
    obs, rew, term, trunc, info \
        = env.step(actions)
```

Lecture A -- Summary

- The single-agent MDP assumes stationarity, full observability, and a unique agent. All three assumptions break down in MAS.
- **Physics analogy:** isolated spin $\rightarrow$ interacting spin system. Mean field ignores agent-agent correlations — a fundamental diagnostic.
- **Nash Equilibrium** = unilateral stability. Not necessarily efficient (Prisoner's Dilemma). Mixed strategy in RPS.
- **PettingZoo:** AEC for sequential turns, Parallel for simultaneous steps. Migrating from Gym = adding one dict layer.